

Polynomially conditioned rational ranks for a real quadrature

Research proof; three independent full mathematical reviews completed

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All polynomial norms below use the uniform probability measure on $[0, 1]$. Write $\phi_j(t) = \sqrt{2j+1}P_j(2t-1)$ for its orthonormal Legendre polynomials. Absolute constants may change between occurrences.

Theorem 1. *For every $m \geq 1$ there exist $K = O((m+1)^4)$ real numbers $0 < b_1 < \dots < b_K < 1$ and strictly increasing rational numbers $0 < r_1 < \dots < r_K < 1$ with common denominator $L = O((m+1)^{160})$ such that*

$$\frac{1}{K} \sum_{q=1}^K b_q^j = \frac{1}{j+1} \quad (0 \leq j \leq m+1), \quad (1)$$

$$\frac{1}{K} \sum_{q=1}^K r_q b_q^j = \frac{1}{j+2} \quad (0 \leq j \leq m). \quad (2)$$

Moreover, $b_{q+1} - b_q \geq c/K$ and $b_1, 1 - b_K \geq c/K$, and $\max_q |r_q - b_q| \leq C(m+1)^{-20}$. The leading constant in this last bound may be made arbitrarily small by choosing the baseline perturbation constant smaller and then increasing the absolute constant in the bound for L . The b_q may be taken arbitrarily close to nodes of a real equal-weight quadrature of degree $4m+2$, provided L is increased; the theorem only asserts the stated polynomial bound.

These are the step-function compatibility equations needed when a second distribution has cumulative probabilities r_q at the first one's atoms b_q . This theorem alone does not establish the full ordering law for two discrete distributions and other continuous dice.

Lemma 1 (Separated equal-weight starting rule). *For every $D \geq 1$ and every integer $K \geq CD^4$, a real uniform quadrature exact through degree D exists at nodes*

$$\left| b_q - \frac{q-1/2}{K} \right| \leq \frac{1}{8K}.$$

Proof. Put $t_q = (q-1/2)/K$ and $f_j(t) = \int_0^t \phi_{j-1}(z) dz$, $1 \leq j \leq D$. The constraints

$$F_j(b) = K^{-1} \sum_q f_j(b_q) - \int_0^1 f_j(t) dt = 0$$

are equivalent to exactness through degree D . We use the elementary Legendre bounds

$$\|\phi_j\|_\infty \leq \sqrt{2j+1}, \quad \|\phi'_j\|_\infty \leq CD^{5/2} \quad (j < D), \quad \sum_{j < D} \phi_j(t)^2 \leq D^2.$$

They follow from $|P_j| \leq 1$ and Markov's derivative inequality. The midpoint error yields $\|F(t)\|_2 \leq CD^3/K^2$. If $A = DF(t)$, then

$$A_{jq} = \phi_{j-1}(t_q)/K, \quad KAA^T = K^{-1} \sum_q \Phi(t_q)\Phi(t_q)^T =: G.$$

The entrywise rectangle estimate and the derivative bound give $\|G - I\|_{\text{op}} \leq CD^4/K$, so $G \geq I/2$ once C is large. The right inverse $B = A^T(AA^T)^{-1}$ consequently satisfies $\|B\|_{2 \rightarrow \infty} \leq 2D$.

On $t + \text{range}(B)$ consider $T(b) = b - BF(b)$. For $\|b - t\|_{\infty} \leq R$, the derivative bounds imply

$$\|(DF(b) - A)v\|_2 \leq CD^3R\|v\|_{\infty}.$$

Since BA is the identity on $\text{range}(B)$, the map T is a contraction there with constant at most CD^4R . Take $R = C'D^4/K^2$, with C' sufficiently large to dominate twice $\|BF(t)\|_{\infty}$. Increasing the absolute lower-bound constant in $K \geq CD^4$ ensures $R \leq 1/(8K)$ and $CD^4R \leq 1/2$. The closed radius- R ball in this affine subspace is preserved by T and complete. Its fixed point gives the desired rule. $\square$

Lemma 2 (Polynomial conditioning of Legendre multiplication). *Let $d \geq 2$, $h(t) = P_d(2t - 1)$, $p \in \mathcal{P}_{d-1}$, and $q \in \mathcal{P}_{d-1}$. Then*

$$\|hq\|_2 \geq cd^{-4}\|q\|_2.$$

If additionally $\deg p \leq m$, $\deg q \leq m - 1$, and $d \in \{m + 1, m + 2\}$, then for $0 < \eta \leq 1$,

$$\|p + \eta hq\|_2 \geq c\eta(m + 1)^{-5}(\|p\|_2^2 + \|q\|_2^2)^{1/2}. \quad (3)$$

Proof. The d zeros z_i of h are simple and interior. The positive Gauss–Legendre weights on $[0, 1]$ satisfy

$$\omega_i = \frac{1}{z_i(1 - z_i)h'(z_i)^2}, \quad \sum_i \omega_i = 1.$$

In particular $|h'(z_i)| \geq 2$. These standard Gaussian identities follow by integrating the squares of the cardinal Lagrange polynomials; see the primary reference <https://dlmf.nist.gov/3.5>. Markov's inequality twice gives $\|h''\|_{\infty} \leq Cd^4$. Therefore, on a radius c_0d^{-4} neighborhood of each zero, $|h'| \geq 1$. The neighborhoods can be chosen disjoint: between two zeros lies a zero of h' , requiring distance at least cd^{-4} from either endpoint under the same derivative bound.

There is exactly one zero of h' between each consecutive pair of zeros of h , and no others. Thus $|h|$ first increases and then decreases on each intervening interval; near the endpoints it is monotone, and $|h(0)| = |h(1)| = 1$. It follows that for $a = c_1d^{-4}$, with sufficiently small absolute $c_1 > 0$,

$$|\{t : |h(t)| < a\}| \leq 2da.$$

The elementary polynomial evaluation bound $\|q\|_{\infty} \leq d\|q\|_2$, obtained by expanding in the ϕ_j , shows that at most $2d^3a \leq 1/2$ of the squared norm of q lies in that set. Hence $\|hq\|_2 \geq a\|q\|_2/\sqrt{2}$.

For (3), put $f = p + \eta hq$. Gaussian quadrature at the zeros of h is exact for p^2 , since $2\deg p \leq 2d - 2$. At those zeros $p = f$, so

$$\|p\|_2^2 = \sum_i \omega_i f(z_i)^2 \leq \|f\|_{\infty}^2 \leq C(m + 1)^2 \|f\|_2^2,$$

as $\deg f \leq 2m + 1$. Thus $\eta\|hq\|_2 \leq C(m + 1)\|f\|_2$. The first assertion gives $\eta\|q\|_2 \leq C(m + 1)^5\|f\|_2$, proving (3). $\square$

Proof of Theorem 1. Take $D = 4m + 2$ and a starting rule b^0 from Lemma 1, with an integer K between CD^4 and $2CD^4$. Let d be the smallest odd integer at least $m + 1$, so $d \in \{m + 1, m + 2\}$, and put

$$h(t) = P_d(2t - 1), \quad \eta = c_*(m + 1)^{-20}, \quad r_q^0 = b_q^0 - \eta h(b_q^0).$$

Here $c_* > 0$ is a sufficiently small fixed absolute constant. The function $t - \eta h(t)$ has derivative at least $3/4$, because $\|h'\|_{\infty} \leq d(d + 1)$. As d is odd, its endpoint values are η and $1 - \eta$. Hence

r^0 is strictly increasing inside $(0, 1)$ and its consecutive gaps are at least c/K . The high-degree starting quadrature and Legendre orthogonality give

$$K^{-1} \sum b_q^{0j} = 1/(j+1), \quad K^{-1} \sum r_q^0 b_q^{0j} = 1/(j+2) \quad (j \leq m).$$

In particular $\sum_q r_q^0 = K/2$.

Choose an even integer L comparable to $C_2(m+1)^{160}$. There exist integers a_q with

$$|a_q/L - r_q^0| \leq 1/L, \quad \sum_q a_q = KL/2 :$$

round all Lr_q^0 down, then add one to as many of them as needed. The discrepancy in their total is an integer in $[0, K)$. Set $r_q = a_q/L$. For large enough C_2 , these ranks remain strictly increasing and interior. Their mean is exactly $1/2$.

It remains to correct b while holding r fixed. Set $f_j(t) = \int_0^t \phi_{j-1}(z) dz$ and $H_j(t) = \int_0^t z \phi_{j-1}(z) dz$. Use the $m+1$ ordinary constraints

$$K^{-1} \sum_q f_j(b_q) = \int_0^1 f_j(t) dt \quad (1 \leq j \leq m+1)$$

and the m constraints

$$K^{-1} \sum_q \{r_q f_j(b_q) - H_j(b_q)\} = \int_0^1 \{t f_j(t) - H_j(t)\} dt \quad (1 \leq j \leq m).$$

Because $\deg H_j \leq m+1$ and the rank means already agree, these are equivalent to (1)–(2). Let $F(b, r)$ be their vector of residuals, and let $A = D_b F(b^0, r^0)$. Its columns, multiplied by K , are evaluations of the $2m+1$ polynomials

$$\phi_0, \dots, \phi_m, \quad -\eta h \phi_0, \dots, -\eta h \phi_{m-1}.$$

Their pairwise products have degree at most $4m+2$, so $G = KAA^T$ is their exact continuous Gram matrix. By (3) and $\eta \asymp (m+1)^{-20}$,

$$G \geq c(m+1)^{-50} I.$$

The evaluation vector has Euclidean norm at most $C(m+1)$. Consequently the right inverse $B = A^T(AA^T)^{-1}$ obeys

$$\|B\|_{2 \rightarrow \infty} \leq C(m+1)^{51}.$$

Since $\|f_j\|_\infty \leq 1$, rounding only the ranks gives $\|F(b^0, r)\|_2 \leq \sqrt{m}/L$.

Use the fixed-inverse map $b \mapsto b - BF(b, r)$ on $b^0 + \text{range}(B)$. For $\|b - b^0\|_\infty \leq R$ and $\|r - r^0\|_\infty \leq 1/L$, its deviation from the baseline derivative satisfies

$$\|(D_b F(b, r) - A)v\|_2 \leq C\{(m+1)^3 R + (m+1)/L\} \|v\|_\infty.$$

Indeed the ordinary row derivatives are ϕ'_j , and the derivative of $(r_q - b_q)\phi_j(b_q)$ in b_q is $-\phi_j(b_q) + (r_q - b_q)\phi'_j(b_q)$; the displayed bound follows from the same elementary Legendre estimates used earlier. Take $R = C_3(m+1)^{52}/L$ large enough to dominate twice the initial Newton displacement. With $L \asymp C_2(m+1)^{160}$ and C_2 sufficiently large, this is a contraction with constant at most $1/2$, preserves the closed radius- R ball, and has $R \leq 1/(16K)$. Its fixed point proves all the desired moments. The latter displacement bound preserves both separation and the endpoint gaps of the starting rule. Finally $|r_q - b_q| \leq \eta + L^{-1} + R \leq 2\eta$ after increasing C_2 . The fixed constant c_* may be chosen arbitrarily small; the other absolute constants then depend on that choice. All constants are absolute. $\square$